1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In NMIBC patients treated with intravesical BCG, TLS abundance and organization predict favorable therapy response only when tumor cells are MSH6-proficient and ARNP/ADNP transcriptional activity in circulating tumor cells (CTCs) is low. MSH6 deficiency and/or high ADNP-driven transcriptional programs in tumor cells drive immune evasion and remodeling of neoantigen landscapes, diminishing the protective effect of TLS. Therefore, TLS alone is an insufficient predictor of BCG response in the context of an MSH6-deficient and/or high-ADNP state. A composite biomarker integrating (i) TLS density/architecture from histology, (ii) MSH6 status (IHC and/or sequencing for MMR defects), and (iii) ADNP ctRNA level from CTCs will more accurately stratify NMIBC patients by BCG response, outperforming TLS alone. Secondary modifiers (PD-L1/CD274 expression, TP53 status, tumor mutational burden, estrogen receptor status, and systemic HDL-C) will modulate the strength of the TLS–MSH6–ADNP relationship and improve predictive performance when incorporated. Predictions include: (a) TLS-positive, MSH6-proficient, ADNP-low patients will show highest BCG response rates and longest relapse-free survival; (b) TLS-positive but MSH6-deficient or ADNP-high patients will exhibit reduced response despite TLS presence; (c) a three-way interaction TLS × MSH6 × ADNP will significantly predict response better than TLS alone; (d) the composite score will generalize across prospective NMIBC cohorts and guide earlier escalation to alternative therapies for non-responders.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	ADNP, MSH6
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

ADNP: ADNP in CTCs) both associate with NMIBC outcomes after BCG, yet some TLS-rich tumors still fail BCG; why does immune organization not always translate to clinical response? 2) Tier 1 paths used: Core Path 1 (NMIBC → BCG → ClinicalTrial → Biomarker) provides the therapeutic/clinical-response context; Core Path 2 (ADNP mRNA detected in CTCs → liquid biopsy biomarkers) supplies a mechanistic readout of tumor-intrinsic transcriptional programs accessible noninvasively; Core Path 3 (MSI/MSH6 and AI-derived signatures) motivates inclusion of mismatch-repair status (MSH6) as a modifier of immune visibility.

MSH6: MSH6 and AI-derived signatures) motivates inclusion of mismatch-repair status (MSH6) as a modifier of immune visibility.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	PD-L1 (CD274)	Supporting modifier from knowledge graph
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	bTMB	Supporting modifier from knowledge graph
Secondary	ER status (ESR1)	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- PD-L1 (CD274): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TP53: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.